

Inside bones

Just like your heart or eyes, your bones are living organs. Bones are made from cells as well as a mix of materials that make them hard and tough, but also slightly springy. Their structure makes them strong enough to support your body, but light enough not to weigh you down.

ALL ABOUT BONES

Although all bones differ in shape and size, they have the same basic structure. A hard, tough outer layer of compact bone surrounds a lighter mass of spongy bone, while a network of blood vessels delivers the essential supplies that keep your bones alive. Some bones also have a central space filled with a jellylike material called bone marrow.

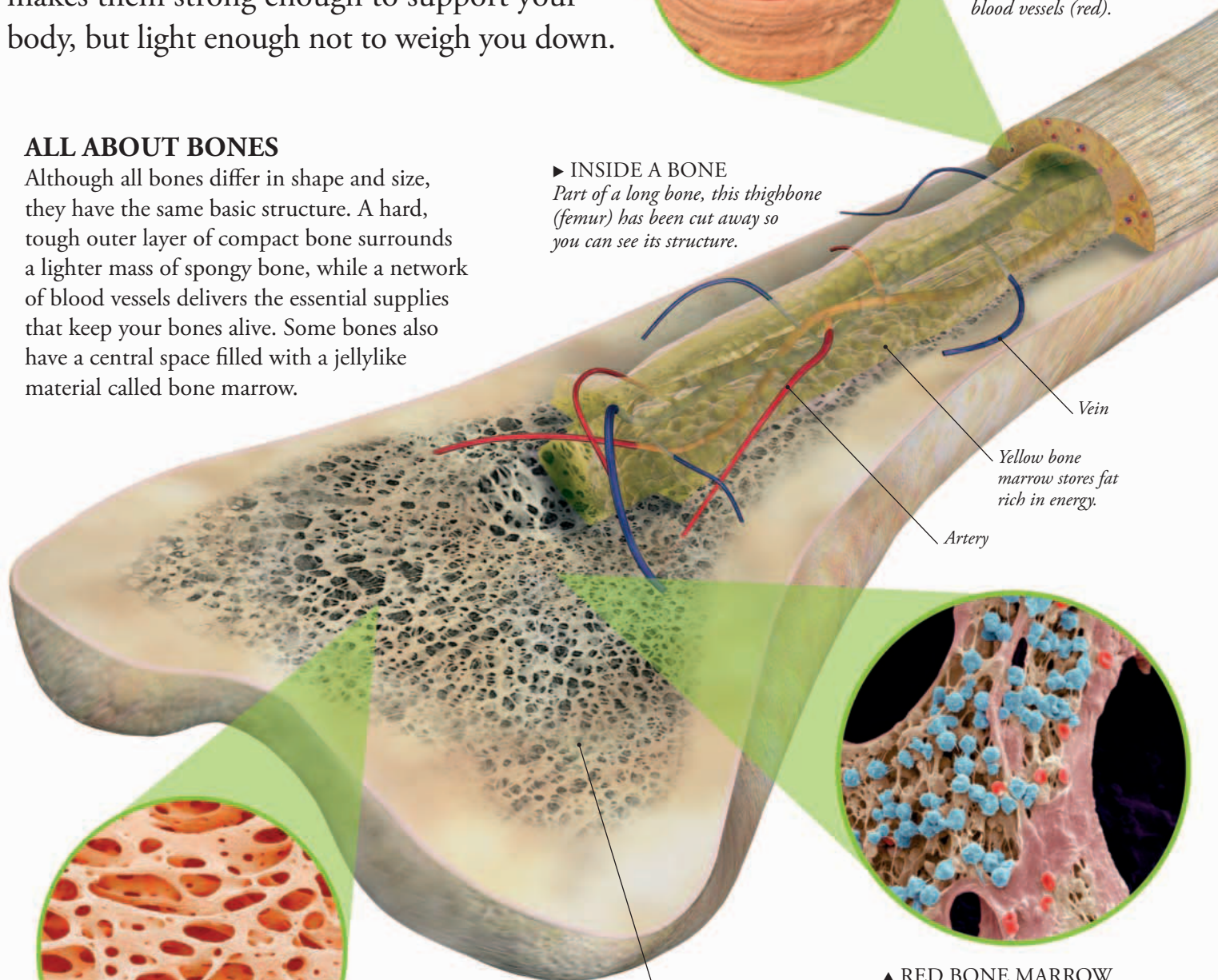


◀ COMPACT BONE

This outer part of the bone is made from cylinders of hard bone tissue that run along the length of the bone, making it strong. At the center are blood vessels (red).

▶ INSIDE A BONE

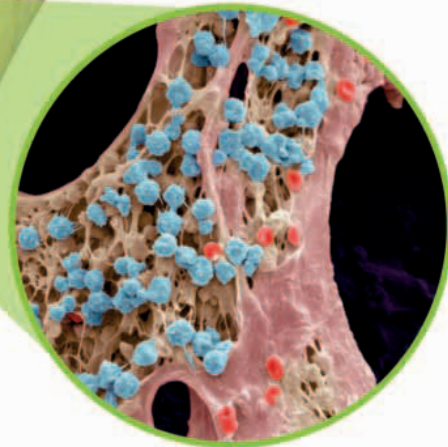
Part of a long bone, this thighbone (femur) has been cut away so you can see its structure.



◀ SPONGY BONE

Despite its name, spongy bone is not soft but strong. Lighter than compact bone, it helps reduce the bone's total weight.

Spongy bone is thickest near the ends of the bone because it is here that your bone has to bear the most stress.



▲ RED BONE MARROW

This soft tissue fills the spaces in spongy bone. It produces all the different types of blood cells. This close-up image shows both red blood cells (red) and white blood cells (blue) in the bone marrow.